

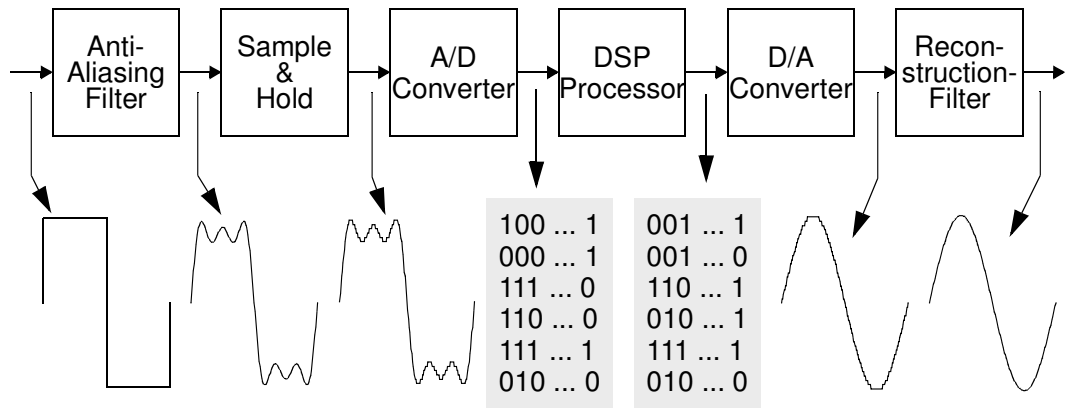
Chapter 3: Sampling and Reconstruction

- 3.1 Introduction
- 3.2 Periodic Sampling
- 3.3 Frequency Domain Analysis (incl. CTFT review)
- 3.4 Relating $X(e^{j\omega})$ to $X_c(j\Omega)$ and $X_s(j\Omega)$
- 3.5 Reconstruction of a Bandlimited Signal from Its Samples
- 3.6 Discrete-Time Processing of Continuous-Time Signals
- 3.7 Changing Sampling Rate using Discrete-Time Processing (Decimation)
- 3.8 Increasing the Sampling Rate (Interpolation) + Rational L/M

3.1 Introduction

- Digital signal processing (DSP) often interfaces with the physical world: continuous-time signals $\rightarrow$ samples $\rightarrow$ digital processing $\rightarrow$ continuous-time output.
- Core questions of this chapter:
 - How does sampling change a signal in time and frequency?
 - When can we perfectly reconstruct the original continuous-time signal?
 - How do we change sampling rates (downsample/upsample) without introducing distortion?
- Vocabulary:
 - Sampling period T (seconds), sampling frequency $f_s = \frac{1}{T}$ (Hz)
 - Sampling radian frequency $\Omega_s = 2\pi f_s = \frac{2\pi}{T}$ (rad/s)

3.1 Introduction



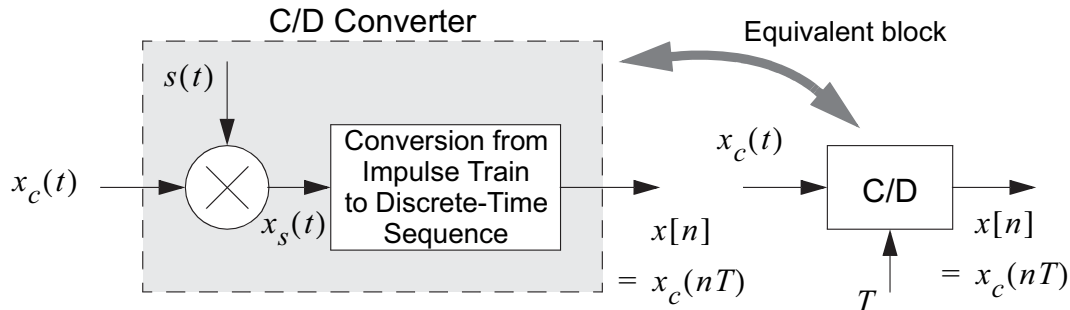
3.2 Periodic Sampling

- We obtain a discrete-time sequence by uniform sampling of a continuous-time signal:

$$x[n] \triangleq x_c(nT), \quad n \in \mathbb{Z}.$$

- The sampling period T sets the sampling rate $f_s = \frac{1}{T}$ and $\Omega_s = \frac{2\pi}{T}$.
- In analysis, it is often useful to separate:
 - the *sequence* $x[n]$ (what a DSP system operates on), and
 - an *impulse-sampled* continuous-time signal $x_s(t)$ (useful for Fourier analysis).

3.2 Periodic Sampling



3.2 Periodic Sampling

Ideal (impulse-train) sampling model.

- Define the sampling impulse train

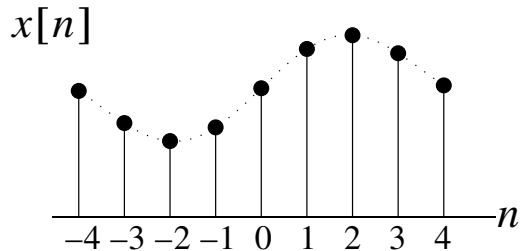
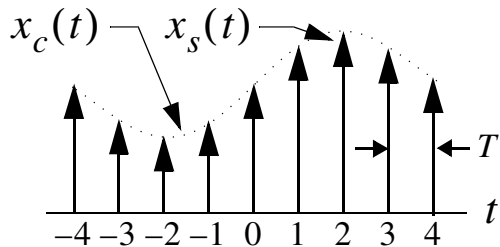
$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT).$$

- The impulse-sampled signal is

$$x_s(t) = x_c(t) s(t) = \sum_{n=-\infty}^{\infty} x_c(nT) \delta(t - nT).$$

- Notice: the impulse weights are exactly the samples $x[n] = x_c(nT)$.

3.2 Periodic Sampling



3.3 Frequency Domain Analysis

CTFT of the sampling impulse train.

- Sampling train: $s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$.
- Its continuous-time Fourier transform (CTFT) is another impulse train:

$$S(j\Omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_s), \quad \Omega_s = \frac{2\pi}{T}.$$

3.3 Frequency Domain Analysis

Spectrum of an impulse-sampled signal.

- $x_s(t) = x_c(t) s(t) \implies$ in frequency:

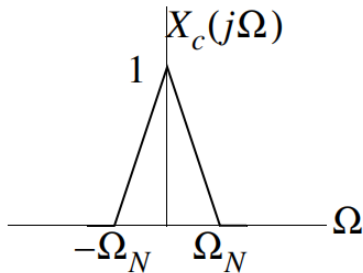
$$X_s(j\Omega) = \frac{1}{2\pi} X_c(j\Omega) * S(j\Omega).$$

- Convolution with an impulse train produces shifted copies (spectral replicas):

$$X_s(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s)).$$

3.3 Frequency Domain Analysis

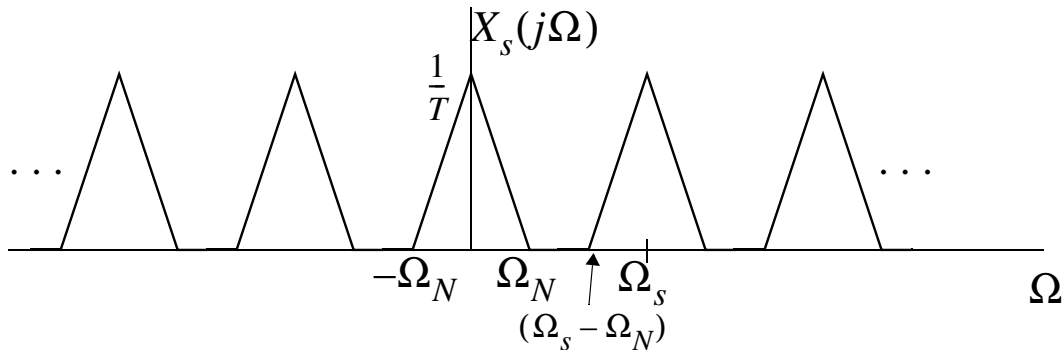
Suppose that $x_c(t)$ has bandlimited Fourier transform as shown below



Fourier transform of $x_c(t)$

3.3 Frequency Domain Analysis

Interpretation: sampling *periodically replicates* the original spectrum every Ω_s .



3.3 Frequency Domain Analysis

Aliasing and the Nyquist condition.

- Let $x_c(t)$ be bandlimited to $|\Omega| \leq \Omega_N$ (i.e., $X_c(j\Omega) = 0$ for $|\Omega| > \Omega_N$).
- Spectral replicas do not overlap if

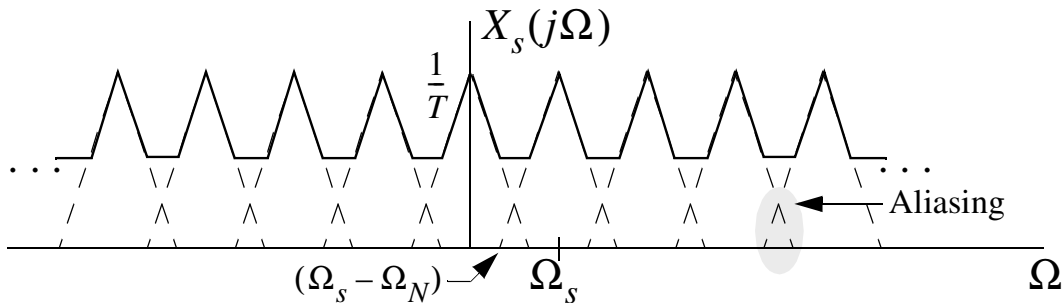
$$\Omega_s > 2\Omega_N \iff f_s > 2f_{\max}.$$

Nyquist Sampling Theorem

If $x_c(t)$ is bandlimited to $|\Omega| \leq \Omega_N$ and $\Omega_s > 2\Omega_N$, then $x_c(t)$ is uniquely determined by its samples $x[n] = x_c(nT)$.

3.3 Frequency Domain Analysis

If $\Omega_s \leq 2\Omega_N$, replicas overlap, different continuous-time frequencies “fold” into the same discrete-time band: this is **aliasing**.



3.3 Frequency Domain Analysis

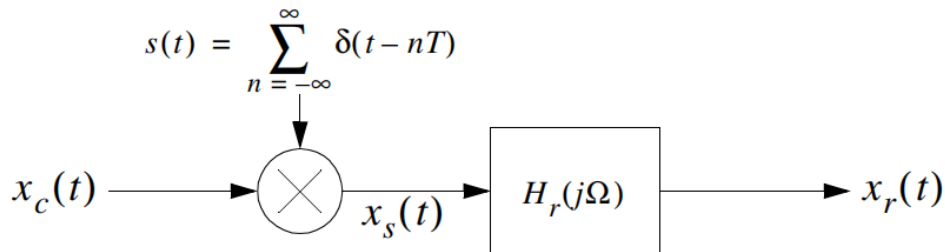
Perfect reconstruction via an ideal lowpass filter.

- If replicas do not overlap, we can extract the baseband copy of $X_c(j\Omega)$ from $X_s(j\Omega)$.
- Choose an *ideal* reconstruction filter

$$H_r(j\Omega) = \begin{cases} T, & |\Omega| \leq \Omega_c, \\ 0, & \text{otherwise,} \end{cases} \quad \Omega_N < \Omega_c < \Omega_s - \Omega_N.$$

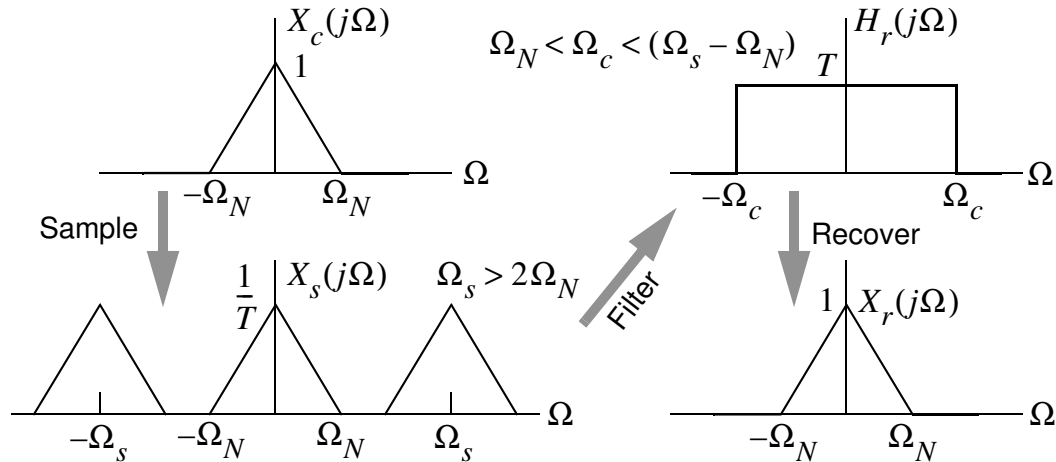
- Then (within the passband) $X_r(j\Omega) = H_r(j\Omega)X_s(j\Omega) = X_c(j\Omega)$, hence $x_r(t) = x_c(t)$.

3.3 Frequency Domain Analysis



Ideal lowpass filter recovery system

3.3 Frequency Domain Analysis



3.3 Frequency Domain Analysis

Example 3.1 (sampled cosine).

Let

$$x_c(t) = \cos(\Omega_0 t).$$

Sampling gives

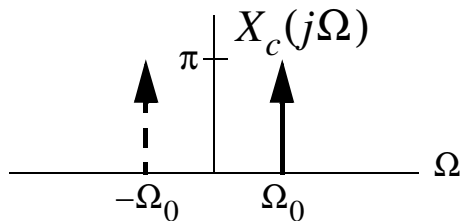
$$x[n] = x_c(nT) = \cos(\Omega_0 nT) = \cos(\omega_0 n), \quad \omega_0 = \Omega_0 T.$$

- Discrete-time frequency is *wrapped* modulo 2π .
- Different Ω_0 can produce the same ω_0 (aliasing in continuous-time frequency).

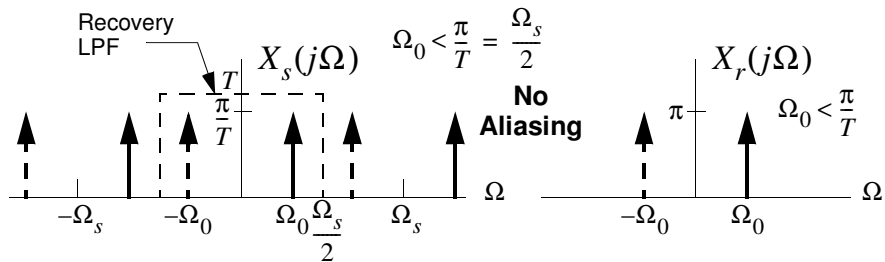
3.3 Frequency Domain Analysis

Suppose that $x_c(t) = \cos \Omega_o t$

The spectrum of $x_c(t)$

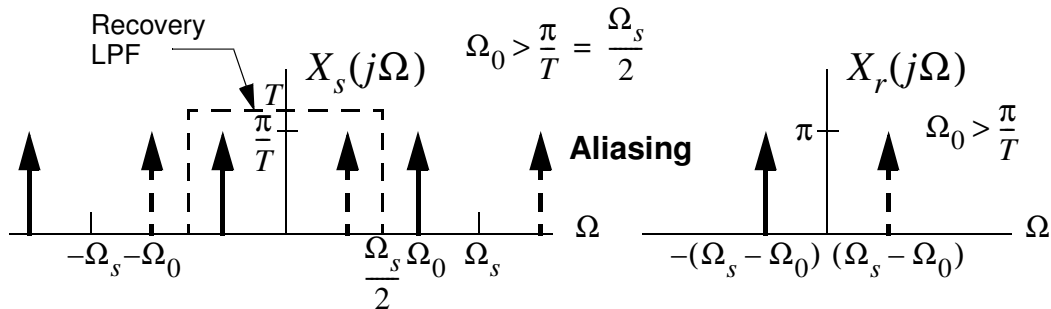


3.3 Frequency Domain Analysis



Pre and Post reconstruction spectra

3.3 Frequency Domain Analysis



Pre and Post reconstruction spectra, with aliasing

3.3 Frequency Domain Analysis

3.3.1 Continuous-Time Fourier Transform (CTFT) review.

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t} dt, \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t} d\Omega.$$

- Ω is continuous-time radian frequency (rad/s).
- Many sampling results follow from: multiplication in time $\leftrightarrow$ convolution in frequency.

3.3 Frequency Domain Analysis

Rectangular pulse and sinc.

$$\text{rect}\left(\frac{t}{\tau}\right) = \begin{cases} 1, & |t| < \frac{\tau}{2}, \\ 0, & \text{otherwise.} \end{cases} \quad \text{sinc}(x) \triangleq \frac{\sin(\pi x)}{\pi x}.$$

A standard transform pair (with the above conventions) is

$$\text{rect}\left(\frac{t}{\tau}\right) \Longleftrightarrow \tau \text{sinc}\left(\frac{\Omega\tau}{2\pi}\right).$$

3.3 Frequency Domain Analysis

Selected CTFT properties (useful in sampling).

- **Time shift:** $x(t - t_0) \iff e^{-j\Omega t_0} X(j\Omega)$
- **Modulation:** $x(t)e^{j\Omega_0 t} \iff X(j(\Omega - \Omega_0))$
- **Convolution:** $x(t) * h(t) \iff X(j\Omega)H(j\Omega)$
- **Multiplication:** $x(t)h(t) \iff \frac{1}{2\pi} X(j\Omega) * H(j\Omega)$
- **Impulse:** $\delta(t) \iff 1$

3.4 Relating $X(e^{j\omega})$ to $X_c(j\Omega)$ and $X_s(j\Omega)$

DTFT definition.

For a sequence $x[n]$:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}, \quad \omega \in \mathbb{R}.$$

Key link to impulse sampling.

$$X_s(j\Omega) = \int_{-\infty}^{\infty} x_s(t) e^{-j\Omega t} dt = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n T}.$$

Therefore

$$X_s(j\Omega) = X(e^{j\omega}) \Big|_{\omega=\Omega T} \iff X(e^{j\omega}) = X_s\left(j\frac{\omega}{T}\right).$$

3.4 Relating $X(e^{j\omega})$ to $X_c(j\Omega)$ and $X_s(j\Omega)$

DTFT in terms of the continuous-time spectrum.

Starting from

$$X_s(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s)),$$

substitute $\Omega = \frac{\omega}{T}$ and $\Omega_s = \frac{2\pi}{T}$:

$$X(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\frac{\omega - 2\pi k}{T}\right).$$

- $X(e^{j\omega})$ is 2π -**periodic** in ω .
- Aliasing is “hidden” inside the sum: overlapping analog replicas add together.

3.5 Reconstruction of a Bandlimited Signal

Reconstruction as filtering of the impulse-sampled signal.

- If $x_s(t) = \sum x[n]\delta(t - nT)$, then reconstruction is

$$x_r(t) = x_s(t) * h_r(t).$$

- For the ideal lowpass reconstruction filter

$$H_r(j\Omega) = \begin{cases} T, & |\Omega| \leq \Omega_c, \\ 0, & \text{otherwise,} \end{cases}$$

the impulse response is

$$h_r(t) = \frac{T}{2\pi} \int_{-\Omega_c}^{\Omega_c} e^{j\Omega t} d\Omega = T \frac{\sin(\Omega_c t)}{\pi t}.$$

3.5 Reconstruction of a Bandlimited Signal

Special (and common) choice: $\Omega_c = \Omega_s/2 = \pi/T$.

Then

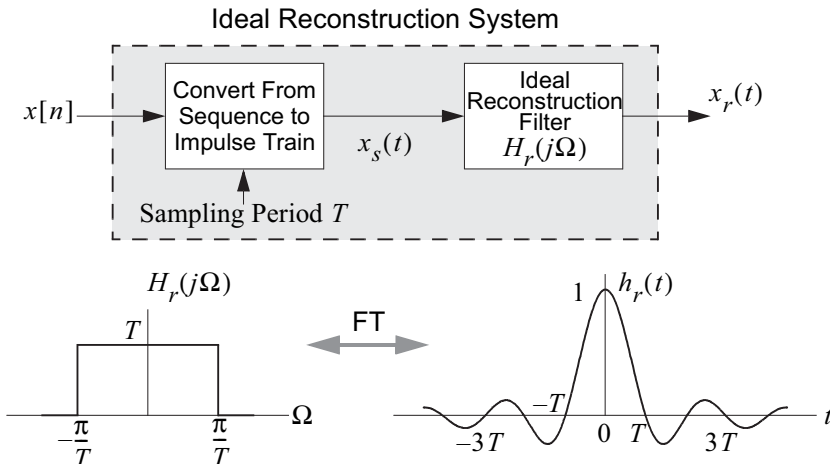
$$h_r(t) = T \frac{\sin(\pi t/T)}{\pi t} = \text{sinc}\left(\frac{t}{T}\right),$$

and the reconstruction (sinc interpolation) formula becomes

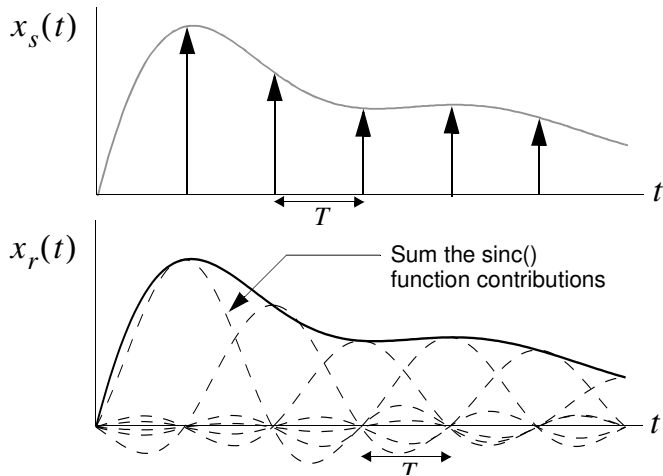
$$x_r(t) = \sum_{n=-\infty}^{\infty} x[n] \text{sinc}\left(\frac{t}{T} - n\right).$$

- This is the classic **Whittaker–Shannon interpolation formula**.
- In practice, we approximate the ideal sinc / ideal LPF with realizable filters.

3.5 Reconstruction of a Bandlimited Signal



3.5 Reconstruction of a Bandlimited Signal



3.5 Reconstruction of a Bandlimited Signal

Two conceptual “conversions”.

- **Continuous-time $\rightarrow$ sequence:** $x[n] = x_c(nT)$
- **Sequence $\rightarrow$ continuous-time:** $x_r(t) = \sum_n x[n] \operatorname{sinc}\left(\frac{t}{T} - n\right)$ (ideal case)

These two blocks are the backbone of mixed-signal DSP systems.

3.6 Discrete-Time Processing of Continuous-Time Signals

The mixed-signal DSP chain.

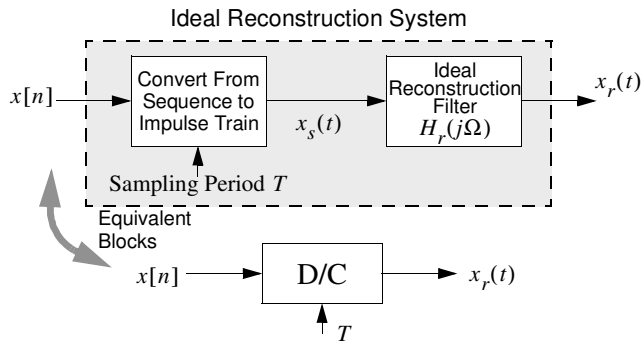
- Sample: $x_c(t) \rightarrow x[n] = x_c(nT)$
- Process digitally: $x[n] \xrightarrow{H(e^{j\omega})} y[n]$
- Reconstruct: $y[n] \rightarrow y_c(t)$ (DAC + reconstruction filter)

If sampling and reconstruction are ideal (and no aliasing occurs), then:

$$Y_c(j\Omega) = H(e^{j\Omega T}) X_c(j\Omega), \quad |\Omega| < \frac{\pi}{T}.$$

So the digital frequency response is “warped” into continuous time by the mapping $\omega = \Omega T$.

3.6 Discrete-Time Processing of Continuous-Time Signals



Ideal D/C reconstruction system

3.6 Discrete-Time Processing of Continuous-Time Signals

Example 3.2 (ideal digital lowpass $\rightarrow$ effective analog lowpass).

If

$$H(e^{j\omega}) = \begin{cases} 1, & |\omega| \leq \omega_c, \\ 0, & \text{otherwise,} \end{cases}$$

then the effective continuous-time passband is

$$|\Omega| \leq \Omega_c, \quad \Omega_c = \frac{\omega_c}{T}.$$

- Fixing ω_c and changing T changes the analog cutoff Ω_c .
- This is a standard way to build tunable continuous-time filters using a fixed digital design.

3.7 Changing Sampling Rate using Discrete-Time Processing

Downsampling / decimation by an integer factor M .

- Define the downsampled sequence

$$x_d[n] = x[nM].$$

- In continuous-time interpretation (if $x[n] = x_c(nT)$):

$$x_d[n] = x_c(nMT)$$

so the sampling period increases by M (sampling rate decreases by M).

Key issue: downsampling can cause *aliasing* unless the sequence is bandlimited (in DTFT) before decimation.

3.7 Changing Sampling Rate using Discrete-Time Processing

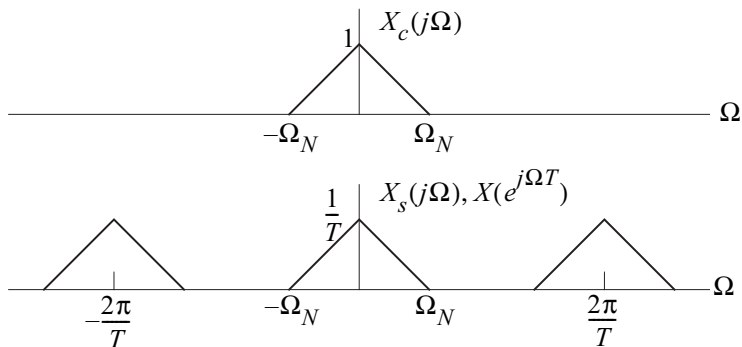
Frequency-domain relationship (decimation).

If $x_d[n] = x[nM]$, then the DTFT satisfies

$$X_d(e^{j\omega}) = \frac{1}{M} \sum_{k=0}^{M-1} X\left(e^{j\left(\frac{\omega}{M} - \frac{2\pi k}{M}\right)}\right).$$

- The spectrum is *compressed* by M and *added* copies appear.
- To avoid aliasing, we typically lowpass-filter $x[n]$ such that $X(e^{j\omega}) = 0$ for $|\omega| > \pi/M$ before downsampling.

3.7 Changing Sampling Rate using Discrete-Time Processing



Spectra of $X_c(j\Omega)$ and $X(e^{j\Omega T})$

3.7 Changing Sampling Rate using Discrete-Time Processing

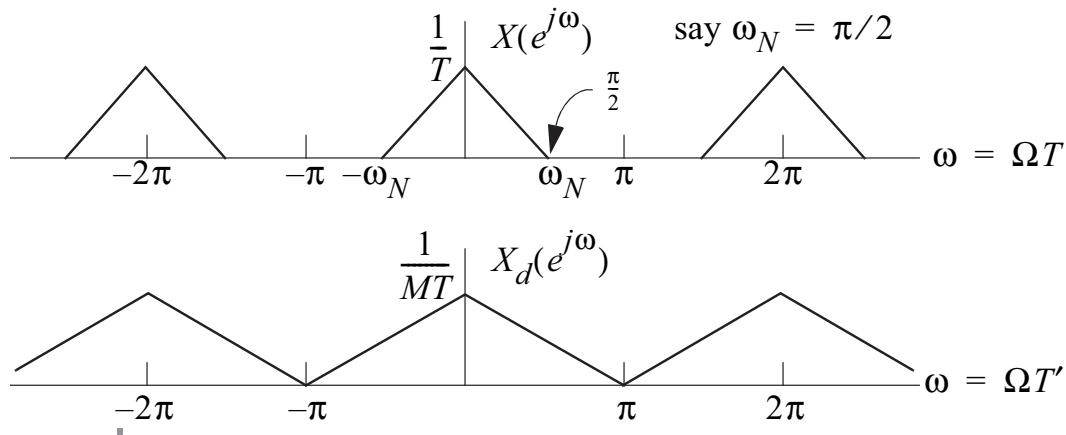
Example 3.7 (decimation by $M = 2$).

For $M = 2$:

$$X_d(e^{j\omega}) = \frac{1}{2} \left[X(e^{j\omega/2}) + X(e^{j(\omega/2 - \pi)}) \right].$$

- If $X(e^{j\omega})$ is confined to $|\omega| \leq \pi/2$, then no overlap occurs $\Rightarrow$ no aliasing.
- Otherwise, the shifted term aliases into baseband.

3.7 Changing Sampling Rate using Discrete-Time Processing



3.7 Changing Sampling Rate using Discrete-Time Processing

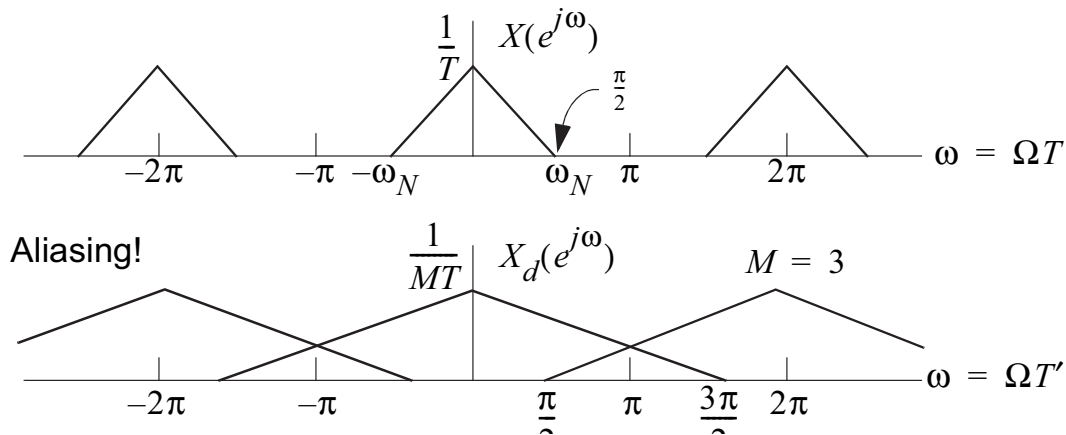
Example 3.8 (decimation by $M = 3$).

For $M = 3$:

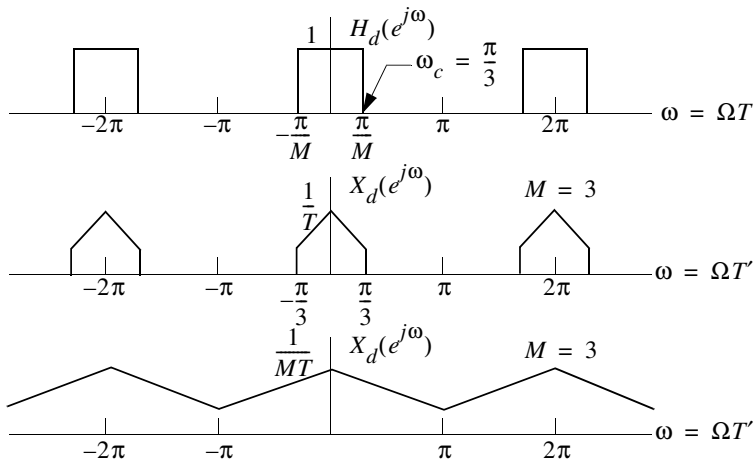
$$X_d(e^{j\omega}) = \frac{1}{3} \left[X(e^{j\omega/3}) + X(e^{j(\omega/3 - 2\pi/3)}) + X(e^{j(\omega/3 - 4\pi/3)}) \right].$$

- Bandlimit before downsampling to $|\omega| \leq \pi/3$ to prevent overlap.
- In practice: **anti-aliasing FIR/IIR** $\rightarrow$ **downsampler**.

3.7 Changing Sampling Rate using Discrete-Time Processing



3.7 Changing Sampling Rate using Discrete-Time Processing



3.8 Increasing the Sampling Rate by an Integer Factor

Upsampling (expansion) by an integer factor L .

- Define the expanded sequence (zero insertion):

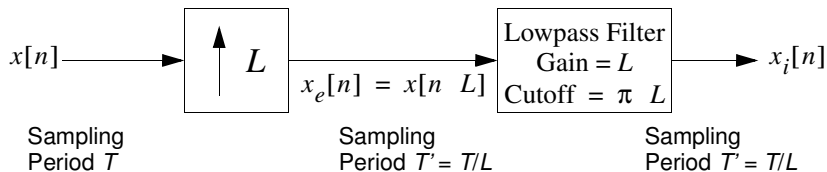
$$x_e[n] = \begin{cases} x[n/L], & n = 0, \pm L, \pm 2L, \dots \\ 0, & \text{otherwise.} \end{cases}$$

- DTFT relationship:

$$X_e(e^{j\omega}) = X(e^{j\omega L}).$$

Interpretation: the spectrum is *compressed* by L and *images* appear.

3.8 Increasing the Sampling Rate by an Integer Factor



3.8 Increasing the Sampling Rate by an Integer Factor

Interpolation filter.

To remove images created by zero insertion, filter $x_e[n]$ with an interpolation lowpass:

$$H_i(e^{j\omega}) = \begin{cases} L, & |\omega| \leq \pi/L, \\ 0, & \text{otherwise.} \end{cases}$$

Its impulse response is

$$h_i[n] = \frac{1}{2\pi} \int_{-\pi/L}^{\pi/L} L e^{j\omega n} d\omega = \text{sinc}\left(\frac{n}{L}\right).$$

The ideal interpolated output can be written as

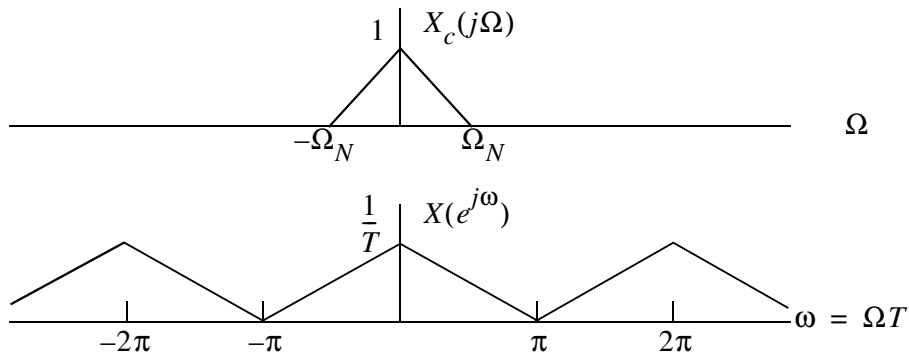
$$x_i[n] = \sum_{k=-\infty}^{\infty} x[k] \text{sinc}\left(\frac{n - kL}{L}\right) = \sum_{k=-\infty}^{\infty} x[k] \text{sinc}\left(\frac{n}{L} - k\right).$$

3.8 Increasing the Sampling Rate by an Integer Factor

Example 3.9 (interpolation by $L = 2$).

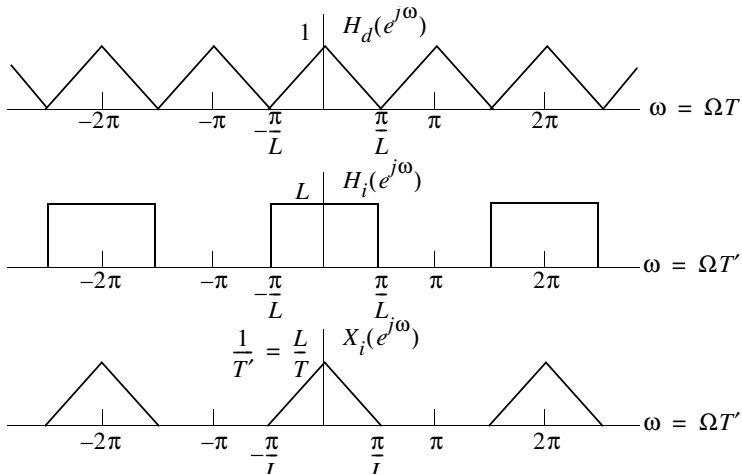
- Insert zeros: $x_e[n] = x[n/2]$ for even n , 0 for odd n .
- The spectrum compresses: $X_e(e^{j\omega}) = X(e^{j2\omega})$, producing an image centered at π .
- A lowpass with cutoff $\pi/2$ removes the image, producing a smooth in-between sample sequence.

3.8 Increasing the Sampling Rate by an Integer Factor



Spectra of $X_c(j\Omega)$ and $X(e^{j\omega})$

3.8 Increasing the Sampling Rate by an Integer Factor



3.8 Increasing the Sampling Rate by an Integer Factor

Linear interpolation (a practical alternative to ideal sinc).

A length- $2L + 1$ triangular impulse response:

$$h_{\text{lin}}[n] = \begin{cases} 1 - \frac{|n|}{L}, & |n| \leq L, \\ 0, & \text{otherwise.} \end{cases}$$

The interpolated output after zero insertion can be written as a convolution:

$$x_i[n] = \sum_{k=-\infty}^{\infty} x_e[k] h_{\text{lin}}[n - k] = \sum_{k=-\infty}^{\infty} x[k] h_{\text{lin}}[n - kL].$$

Its frequency response (magnitude) is approximately lowpass:

$$|H_{\text{lin}}(e^{j\omega})| = \frac{1}{L} \left(\frac{\sin(\omega L/2)}{\sin(\omega/2)} \right)^2.$$

(The phase is approximately linear and depends on the indexing choice.)

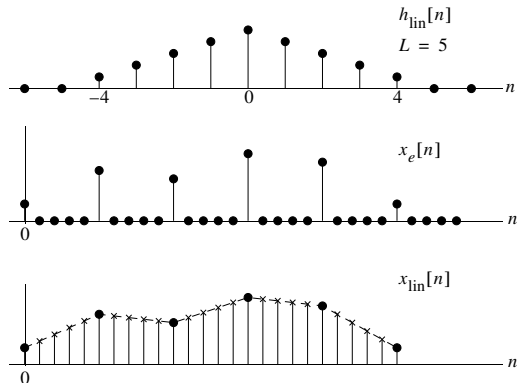
- Cheaper than sinc interpolation, but with more passband droop and less stopband attenuation.

3.8 Increasing the Sampling Rate by an Integer Factor

Example 3.10 (linear interpolation, $L = 5$).

- Insert 4 zeros between samples, then apply the triangular filter.
- Time-domain effect: creates piecewise-linear segments between original sample points.
- Frequency-domain effect: smooths images, but not as aggressively as an ideal lowpass.

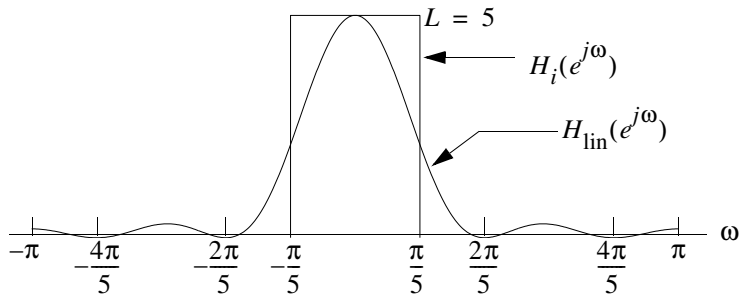
3.8 Increasing the Sampling Rate by an Integer Factor



Linear interpolation; impulse response, & I/O

3.8 Increasing the Sampling Rate by an Integer Factor

$$H_{\text{lin}}(e^{j\omega}) = \frac{1}{L} \left[\frac{\sin(\omega L/2)}{\sin(\omega/2)} \right]^2$$



3.8 Increasing the Sampling Rate by an Integer Factor

3.9.1 Changing the sampling rate by a rational factor L/M .

A common structure:

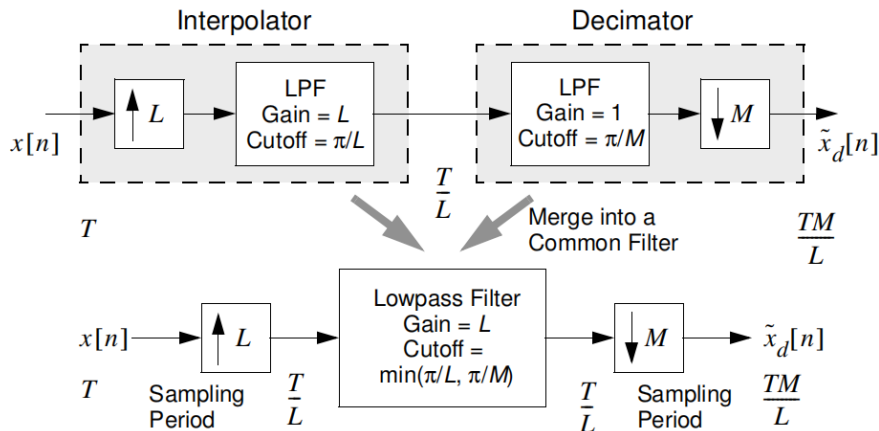
$$x[n] \xrightarrow{\uparrow L} x_e[n] \xrightarrow{H(e^{j\omega})} v[n] \xrightarrow{\downarrow M} y[n].$$

- After upsampling, suppress images with cutoff π/L .
- Before downsampling by M , also ensure the spectrum is within $|\omega| \leq \pi/M$ to avoid aliasing.
- In practice, the image-suppression (post- $\uparrow L$) and anti-alias (pre- $\downarrow M$) constraints can be met with a single lowpass stage.
- Therefore choose the interpolation/anti-alias filter cutoff:

$$\omega_c \leq \min\left(\frac{\pi}{L}, \frac{\pi}{M}\right),$$

with gain typically L .

3.8 Increasing the Sampling Rate by an Integer Factor



3.8 Increasing the Sampling Rate by an Integer Factor

Example 3.11 (CD player oversampling intuition).

- CD audio: $f_s = 44.1$ kHz, desired baseband ≈ 20 kHz.
- A practical DAC includes a **zero-order hold (ZOH)** with impulse response

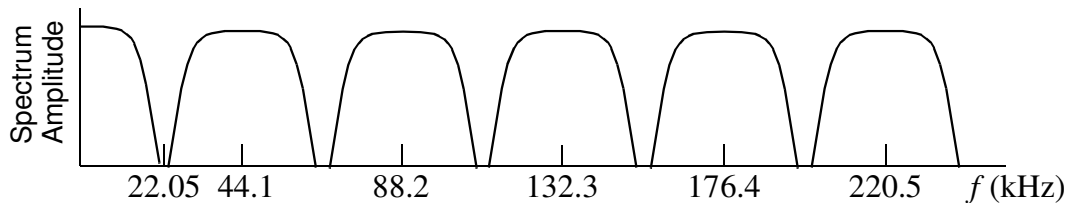
$$h_1(t) = u(t) - u(t - T).$$

Its frequency response is

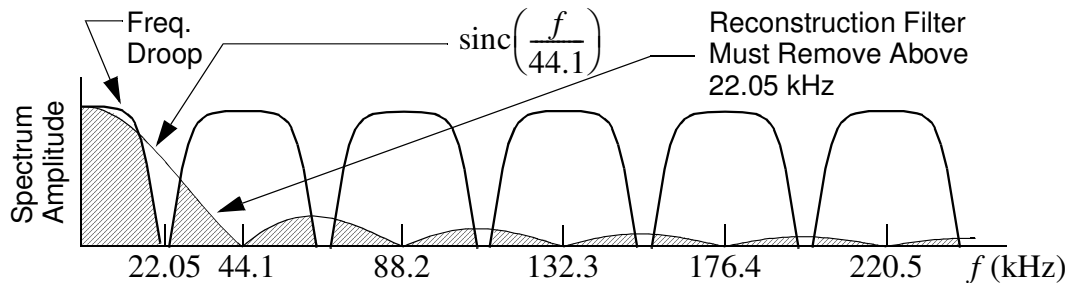
$$H_1(j\Omega) = T e^{-j\Omega T/2} \frac{\sin(\Omega T/2)}{\Omega T/2} \implies |H_1(j2\pi f)| = T \operatorname{sinc}\left(\frac{f}{f_s}\right).$$

- Without oversampling, the analog reconstruction LPF must transition from 20 kHz to 22.05 kHz — tight and expensive.
- Oversampling (e.g., $4\times$: 44.1 kHz $\rightarrow$ 176.4 kHz) pushes images farther away, making the analog LPF gentler and reducing audible ZOH droop.

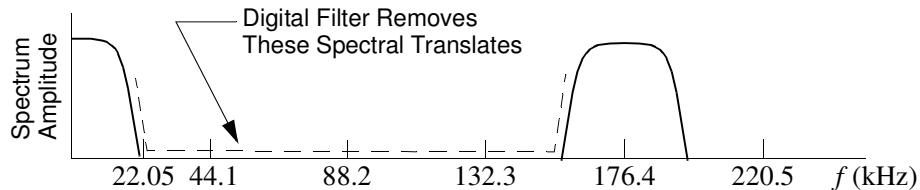
3.8 Increasing the Sampling Rate by an Integer Factor



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